Math 55—Fall 2025—Haiman
Homework 2
Due on Gradescope Wednesday, Sept. 10, 8pm

Problems from Hutchings

Appendix A Exercises (page 27):

1.
5.
7. (as always, justify your answer to the yes/no second part of the question)

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1. List separately the elements and the subsets of $\{\{1,\{2\}\},\{3\}\}$. (There are 2 elements and 4 subsets.)

Elements: $\{1,\{2\}\}, \{3\}$.
Subsets: $\emptyset$, $\{1,\{2\}\}, \{3\}, \{\{1,\{2\}\},\{3\}\}$.

5. Which of the following statements are true, and which are false? Why?

(a) $\{\{\emptyset\}\} \cup \emptyset = \{\emptyset, \{\emptyset\}\}$
(b) $\{\{\emptyset\}\} \cup \{\emptyset\} = \{\emptyset, \{\emptyset\}\}$
(c) $\{\emptyset, \{\emptyset\}\} \cap \{\{\emptyset\}, \{\{\emptyset\}\}\} = \{\emptyset\}$
(d) $\{\emptyset, \{\emptyset\}\} \cap \{\{\emptyset\}, \{\{\emptyset\}\}\} = \{\{\emptyset\}\}$

(a) False. Reasons:
$\emptyset$ means zero element, i.e. empty set.
$\Rightarrow \{\{\emptyset\}\} \cup \emptyset = \{\{\emptyset\}\}$.

(b) True. Reasons:
The elements of $\{\{\emptyset\}\}$ and $\{\emptyset\}$ are $\{\emptyset\}$ and $\emptyset$ respectively
$\Rightarrow \{\{\emptyset\}\} \cup \{\emptyset\} = \{\emptyset, \{\emptyset\}\}$.

(c) True. Reasons:
Elements of $\{\emptyset, \{\emptyset\}\}$: $\emptyset, \{\emptyset\}$.
Elements of $\{\{\emptyset\}, \{\{\emptyset\}\}\}$: $\{\emptyset\}, \{\{\emptyset\}\}$.
$\Rightarrow$ intersection $= \{\emptyset\}$.

(d) False. Reasons:
The intersection of two sets should be elements.
According to (c), the intersection should be the element $\{\emptyset\}$.

7. Show that $A \cap (B - C) = (A \cap B) - (A \cap C)$. Is it always true that $A \cup (B - C) = (A \cup B) - (A \cup C)$?

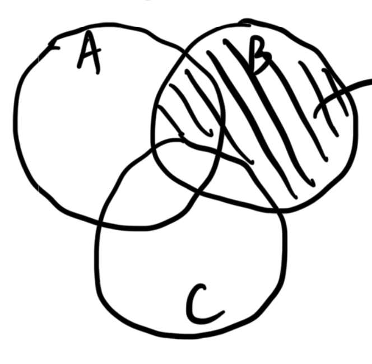
- To show $A \cap (B-C) = (A \cap B) - (A \cap C)$:

① First show $A \cap (B-C) \subseteq (A \cap B) - (A \cap C)$:
Assume $x \in A \cap (B-C)$, then $x \in A$, $x \in (B-C)$
$\Rightarrow x \in B$.
$\Rightarrow x \in A \cap B \Rightarrow x \in (A \cap B) - (A \cap C)$
$\Rightarrow A \cap (B-C) \subseteq (A \cap B) - (A \cap C)$.

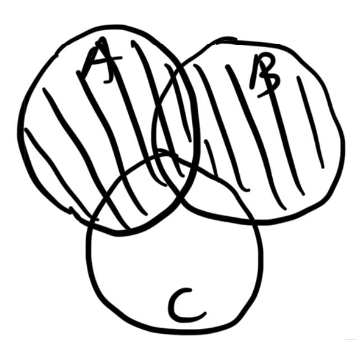
② Then show $(A \cap B) - (A \cap C) \subseteq A \cap (B-C)$:
Assume $x \in (A \cap B) - (A \cap C)$, then $x \in A \cap B$.
$\Rightarrow x \in A$, $x \in B$. $\Rightarrow x \in B - C$
$\Rightarrow x \in A \cap (B-C)$.
$\Rightarrow (A \cap B) - (A \cap C) \subseteq A \cap (B-C)$.
Thus, $(A \cap B) - (A \cap C) = A \cap (B-C)$.

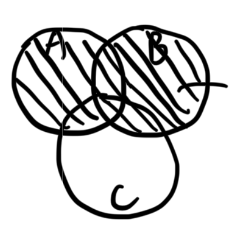
- Is it always true that $A \cup (B-C) = (A \cup B) - (A \cup C)$?

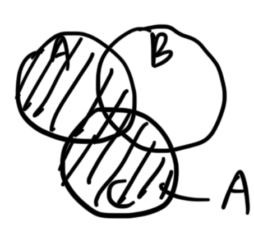
No.



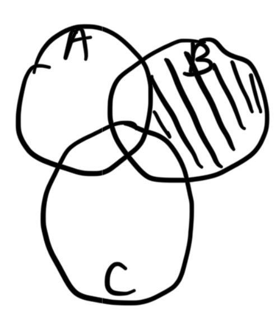
$\Rightarrow A \cup (B-C)$:







$\Rightarrow (A \cup B) - (A \cup C)$:



$\Rightarrow (A \cup B) - (A \cup C) \subseteq A \cup (B-C)$.

Problems from Rosen

Section 2.1

20.
24. (prove or find a counterexample)
28.
32.

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20. Find two sets A and B such that $A \in B$ and $A \subseteq B$.

Take $A = \{1\}$, $B = \{\{1\}, 1\}$.

24. Can you conclude that $A = B$ if A and B are two sets with the same power set?

Yes. Prove ;
If A and B are two sets with the same power set, then A and B have the same subset.
$\Rightarrow$ A and B have the same elements
$\Rightarrow A = B$.

28. Show that if $A \subseteq C$ and $B \subseteq D$, then $A \times B \subseteq C \times D$

Assume $(a,b) \in A \times B$.
then we know $a \in A$, $b \in B$.
$\because A \subseteq C, B \subseteq D$
$\therefore a \subseteq C, b \subseteq D$
$\therefore (a,b) \in C \times D$
Thus, $A \times B \subseteq C \times D$.

32. Suppose that $A \times B = \emptyset$, where A and B are sets. What can you conclude?

At least one of A and B is an empty set. Reasons:
Suppose $A \neq \emptyset$, $B \neq \emptyset$.
then $\exists\, a, b$, s.t. $a \in A$, $b \in B$.
$\Rightarrow (a,b) \in A \times B$.
$\Rightarrow A \times B \neq \emptyset$. contradicts.
Thus, at least one of A and B is an empty set.

Section 2.2

14.
32. (justify your answers)
54.

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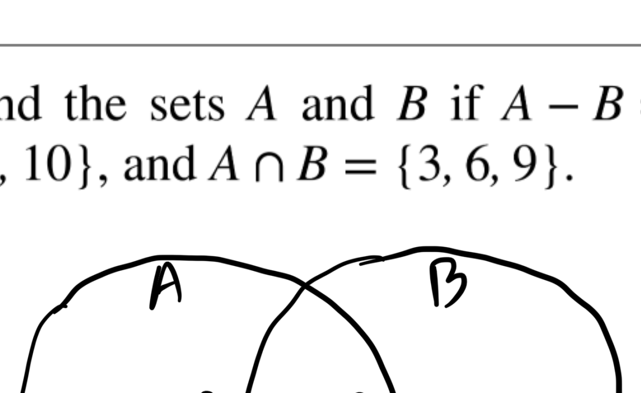
14. Find the sets A and B if $A - B = \{1, 5, 7, 8\}$, $B - A = \{2, 10\}$, and $A \cap B = \{3, 6, 9\}$.



$\Rightarrow A = \{1,3,5,6,7,8,9\}$
$B = \{2,3,6,9,10\}$.

32. Can you conclude that $A = B$ if A, B, and C are sets such that

a) $A \cup C = B \cup C$? **b)** $A \cap C = B \cap C$?
c) $A \cup C = B \cup C$ and $A \cap C = B \cap C$?

a) No. Counter-example:
Take $A = \{4\}$, $B = \{5\}$, $C = \{4,5\}$.
$\Rightarrow A \cup C = \{4,5\}$, $B \cup C = \{4,5\}$.
$\Rightarrow A \cup C = B \cup C$ but $A \neq B$.

b) No.
Take $A = \{1,2,3\}$, $B = \{3,4,5\}$, $C = \{3\}$.
$\Rightarrow A \cap C = B \cap C = \{3\}$. but $A \neq B$.

c) Yes. Prove:
① if $x \in C$.
then $A \cap C = B \cap C$
$\Rightarrow x \in A \Leftrightarrow x \in B$.
$\Rightarrow A = B$
② if $x \notin C$:
1°. For $\forall x \in A$, $x \in (A \cup C)$.
$\Rightarrow x \in B \cup C$.
$\because x \notin C$
$\therefore x \in B$. $\Rightarrow A \subseteq B$
2°. For $\forall x \in B$, $x \in (B \cup C)$
$\Rightarrow x \in A \cup C$
$\because x \notin C$
$\therefore x \in A$ $\Rightarrow B \subseteq A$
$\Rightarrow A = B$. #.

54. Let $A_i = \{\ldots, -2, -1, 0, 1, \ldots, i\}$. Find

a) $\bigcup_{i=1}^{n} A_i$. **b)** $\bigcap_{i=1}^{n} A_i$.

a) $\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \ldots \cup A_i$
$= \{\ldots, -2, -1, 0, 1\} \cup \{\ldots, -2, -1, 0, 1, 2\}$
$\cup \ldots \cup \{\ldots, -2, -1, 0, 1 \ldots i\} = \{\ldots, -2, -1, 0, 1, \ldots i\}$.
$= A_i$

b) $\bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap A_3 \ldots \cap A_i$
$= \{\ldots, -2, -1, 0, 1\} \cap \{\ldots, -2, -1, 0, 1, 2\}$
$\cap \ldots \cap \{\ldots, -2, -1, 0, 1 \ldots i\} = \{\ldots, -2, -1, 0, 1\} = A_1$